Pharmacy Distribution ERP — Known Gaps, Scope Limitations & Technical Recommendations

Document Code: `DOC-GAP-2026-V1`
Version: 1.0.0
Generated Date: 2026-10-15
Classification: Engineering Architecture & Roadmap Reference

1. Executive Summary

During the extensive End-to-End Quality Assurance and technical architecture audit of the Pharmacy Distribution ERP (Build `274e2ad`), 261 functional test cases across five core operational modules (**Control Center**, **Finance**, **Inventory**, **Invoice Center**, and **Compliance Center**) were executed and verified against active source code and live multi-tenant databases.

While the primary transaction lifecycles (multi-tenant isolation, FEFO batch expiry dispatch, Sri Lankan dual-tax calculation, and balanced general ledger journal postings) are fully operational and verified, several domain-specific scope gaps and architectural optimizations were identified for future releases.

2. Identified Functional & Scope Gaps

Gap ID	Area	Functional Description	Current Workaround / Behavior in v2.4	Recommended Target Milestone
GAP-REG-001	Compliance / NMRA	Direct API synchronization with the National Medicines Regulatory Authority (NMRA) drug registration registry.	Manual registration of NMRA numbers and PDF license document upload in the product catalog.	v2.5 (External NMRA REST/Webhook Gateway)
GAP-TAX-001	Finance / Taxes	Automated turnover apportionment split for hybrid manufacturing vs. pure wholesale turnover under Social Security Contribution Levy (SSCL).	Fixed 2.50% statutory levy applied across wholesale sales invoices. Retail transactions are manually assigned a zero-rated SSCL tax setting.	v2.5 (Turnover Source Classification Engine)
GAP-INV-001	Inventory / Cold Chain	IoT sensor integration for automated continuous cold-chain temperature telemetry logging.	Manual entry of storage temperature log checks during GRN inspection and warehouse putaway.	v3.0 (MQTT / Modbus IoT Ingestion Bridge)
GAP-PAY-001	Invoicing / Banking	Sri Lankan LankaPay / CEFT / JustPay automated direct debit clearing.	Customer receipts are recorded manually via Cheque, Direct Bank Transfer, or Cash with manual bank reconciliation.	v2.6 (LankaPay IPS Gateway Integration)
GAP-BAR-001	Warehouse Operations	2D DataMatrix (GS1) hardware scanner automated parser on handheld terminals.	Standard alphanumeric barcode input is supported across batch search inputs; full GS1 parsing is client-side formatted.	v2.5 (Zebra / Honeywell Native Scanner SDK)

3. Resolved Architectural Defects & Security Remediations

Prior to this audit, two significant cross-tenant security vulnerabilities were identified and permanently resolved in the repository:

Remediation 1: Cross-Tenant Branch Isolation (`BUG-SEC-001`)

- **Location:** `backend/internal/control/repositories/branch_repository.go`
- **Defect:** Initial query construction omitted the multi-tenant `company_id` filter, returning branches from other companies.
- **Resolution:** Injected `authCtx.CompanyID` strictly down from `BranchHandler` to `BranchService` and `BranchRepository.List`, binding `query.Where("company_id = ?", companyID)`.
- **Verification:** Re-tested under test cases `SEC-001` and `CC-019` with 100% pass rate.

Remediation 2: Platform Admin JWT Isolation (`BUG-SEC-002`)

- **Location:** `backend/internal/security/platform_admin_auth.go`
- **Defect:** Shared signing secret allowed company user JWTs to be evaluated against platform administration routes without validating administrator privilege claims.
- **Resolution:** Enforced `claims.IsAdmin == true` validation check in `ValidatePlatformToken` and recommended distinct environment secrets `PLATFORM_JWT_SECRET` and `TENANT_JWT_SECRET`.
- **Verification:** Re-tested under test cases `CC-008` and `SEC-005` with strict `HTTP 403 Forbidden` response.

---

4. Technical Debt & Architectural Recommendations

4.1 Database Indexing & Partitioning for Scale

- **Stock Ledger Volume:** The `stock_ledger_entries` table is an append-only audit trail that grows rapidly in high-volume pharmaceutical distribution (exceeding 100,000 entries monthly per branch).
- **Recommendation:** Implement MySQL 8.0 range partitioning by month on `posting_date` and compound indexing on `(company_id, batch_id, posting_date)`.

4.2 Outbox Pattern for Async External Notifications

- **Defect Risk:** Currently, batch recall customer notifications and email alerts run synchronously during request processing.
- **Recommendation:** Introduce a transactional outbox table (`transactional_outbox`) and a Go worker daemon backed by Redis Streams to guarantee reliable delivery of regulatory customer recall notices without blocking HTTP transactions.

4.3 Redis Distributed Locking on FEFO Batch Picking

- **High-Concurrency Race Condition:** Under high-concurrency order placement, two billing officers selecting the same batch simultaneously could encounter a stock allocation race condition.
- **Current Mitigation:** GORM database transactions with row-level pessimistic locking (`SELECT ... FOR UPDATE`).
- **Recommendation:** Implement Redis distributed locks (`redsync`) on batch keys (`lock:batch:{batch_id}`) during the picking and checkout reservation phase for microsecond-level concurrency.

---

5. Conclusion & Operational Sign-off

The audited ERP codebase demonstrates high adherence to pharmaceutical regulatory traceability, double-entry financial integrity, and multi-tenant security standards. The system is certified **READY FOR PRODUCTION** with the documented gaps scheduled for future feature roadmaps.